KAIZEN MATHS

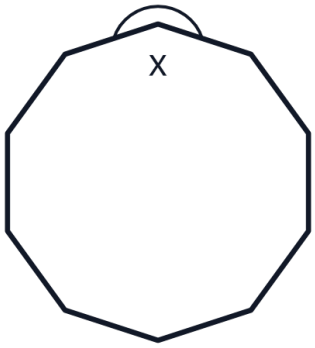
# Lesson starter

**Polygons and Interior/Exterior Angles**

Regular Polygons

Name _____

**SA1**




Find one interior angle of a regular decagon.

GEO-BAE7-L2 · SA

Show your working

KAIZEN MATHS

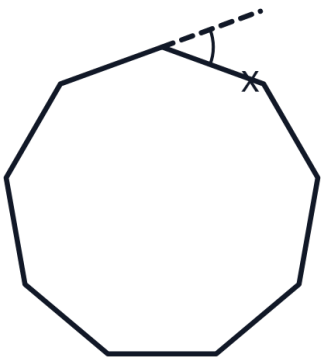
Lesson starter

**Polygons and Interior/Exterior Angles**

Regular Polygons

Name _____

**SB1**




Find one exterior angle of a regular nonagon.

GEO-BAE7-L2 · SB

Show your working

KAIZEN MATHS

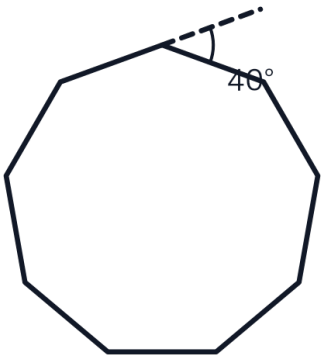
# Lesson starter

**Polygons and Interior/Exterior Angles**

Regular Polygons

Name _____

SC1




Each exterior angle of a regular polygon is 40°. Find the number of sides.



Show your working

KAIZEN MATHS

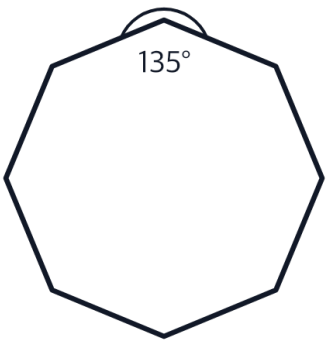
Lesson starter

Polygons and Interior/Exterior Angles

Regular Polygons

Name _____

SD1




Each interior angle of a regular polygon is 135°. Find the number of sides.



Show your working

KAIZEN MATHS

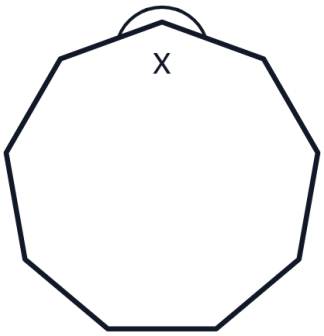
# Exit ticket

**Polygons and Interior/Exterior Angles**

Regular Polygons

Name _____

**EA1**




Find one interior angle of a regular nonagon.



Show your working

KAIZEN MATHS

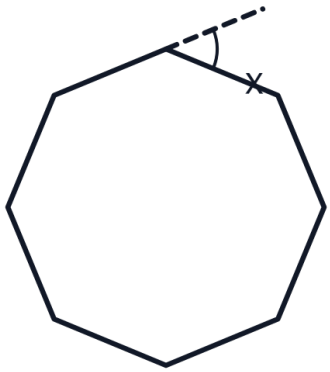
# Exit ticket

**Polygons and Interior/Exterior Angles**

Regular Polygons

Name _____

**EB1**




Find one exterior angle of a regular octagon.



Show your working

KAIZEN MATHS

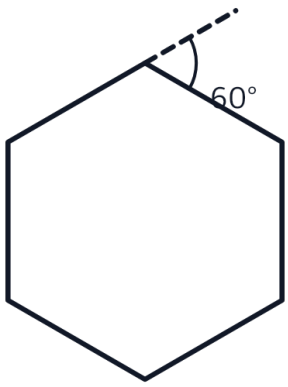
# Exit ticket

**Polygons and Interior/Exterior Angles**

Regular Polygons

Name _____

**EC1**




Each exterior angle of a regular polygon is 60°.
Find the number of sides.



Show your working

KAIZEN MATHS

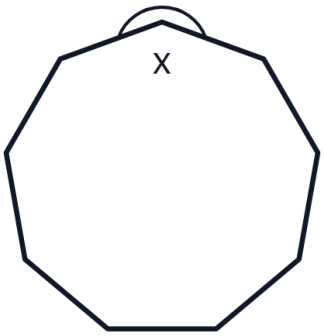
# Exit ticket

**Polygons and Interior/Exterior Angles**

Regular Polygons

Name _____

**EA1**




Find one interior angle of a regular nonagon.



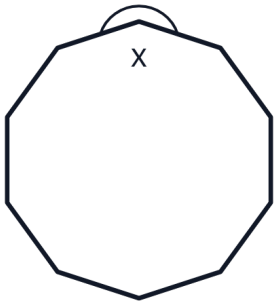
Show your working

# Teacher answers & working

**Polygons and Interior/Exterior Angles · Regular Polygons**

GEO-BAE7-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## SA1




Find one interior angle of a regular decagon.

**Answer:**

144°

**Working:**

A regular decagon has 10 equal exterior angles.

Each exterior angle = 360° ÷ 10 = 36°.

Interior angle + exterior angle = 180°.

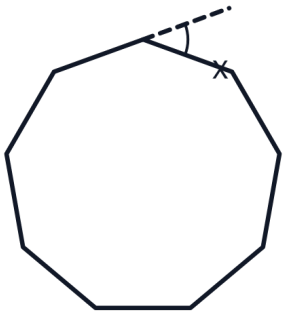
Interior angle = 180° - 36° = 144°.

# Teacher answers & working

**Polygons and Interior/Exterior Angles · Regular Polygons**

GEO-BAE7-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## SB1



Find one exterior angle of a regular nonagon.

**Answer:**

40°

**Working:**

The exterior angles of a polygon add to 360°.

In a regular nonagon, the 9 exterior angles are equal.

Each exterior angle = 360° ÷ 9.

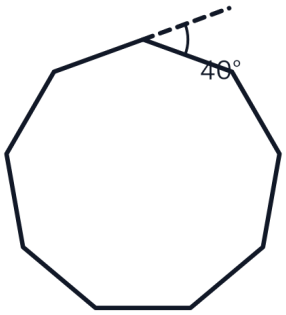
Each exterior angle = 40°.

# Teacher answers & working

**Polygons and Interior/Exterior Angles · Regular Polygons**

GEO-BAE7-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## SC1




Each exterior angle of a regular polygon is 40°. Find the number of sides.

**Answer:**

9

**Working:**

The exterior angles of any polygon add to 360°.

In a regular polygon, all exterior angles are equal.

Number of sides = 360° ÷ exterior angle.

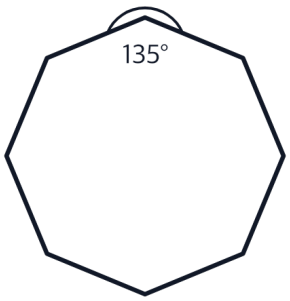
Number of sides = 360° ÷ 40° = 9.

# Teacher answers & working

**Polygons and Interior/Exterior Angles · Regular Polygons**

GEO-BAE7-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

**SD1**




Each interior angle of a regular polygon is 135°. Find the number of sides.

**Answer:**

8

**Working:**

Interior angle + exterior angle = 180°.

Exterior angle = 180° - 135° = 45°.

Number of sides = 360° ÷ exterior angle.

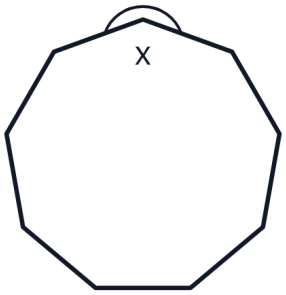
Number of sides = 360° ÷ 45° = 8.

# Teacher answers & working

**Polygons and Interior/Exterior Angles · Regular Polygons**

GEO-BAE7-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## EA1




Find one interior angle of a regular nonagon.

**Answer:**

140°

**Working:**

A regular nonagon has 9 equal exterior angles.

Each exterior angle = 360° ÷ 9 = 40°.

Interior angle + exterior angle = 180°.

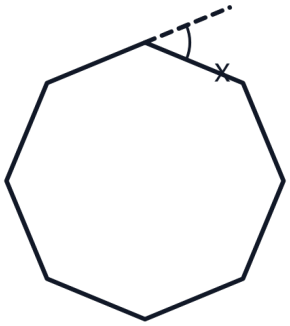
Interior angle = 180° - 40° = 140°.

# Teacher answers & working

**Polygons and Interior/Exterior Angles · Regular Polygons**

GEO-BAE7-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## EB1




Find one exterior angle of a regular octagon.

**Answer:**

45°

**Working:**

The exterior angles of a polygon add to 360°.

In a regular octagon, the 8 exterior angles are equal.

Each exterior angle = 360° ÷ 8.

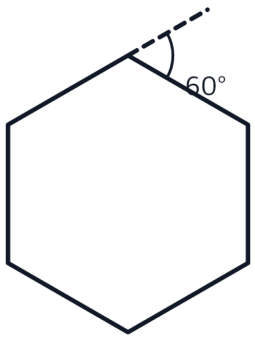
Each exterior angle = 45°.

# Teacher answers & working

**Polygons and Interior/Exterior Angles · Regular Polygons**

GEO-BAE7-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## EC1




Each exterior angle of a regular polygon is 60°. Find the number of sides.

**Answer:**

6

**Working:**

The exterior angles of any polygon add to 360°.

In a regular polygon, all exterior angles are equal.

Number of sides = 360° ÷ exterior angle.

Number of sides = 360° ÷ 60° = 6.